

Databases for Big Data

Exercises: Vectorization

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Example database Sales

The following database contains two entity tables, Customer and Product, which store descriptive information about business entities. A Sales relation links these entities and records measurable events (e.g., quantities sold), forming a simple OLAP schema suitable for analytical queries.

```
CREATE TABLE Customer (
  cid    INT PRIMARY KEY,
  name   TEXT NOT NULL,
  region TEXT
);
```

```
CREATE TABLE Product (
  pid     INT PRIMARY KEY,
  name    TEXT NOT NULL,
  category TEXT
);
```

```
CREATE TABLE Sales (
  sale_id INT PRIMARY KEY,
  cid     INT NOT NULL,
  pid     INT NOT NULL,
  quantity INT NOT NULL,
  FOREIGN KEY (cid) REFERENCES Customer(cid),
  FOREIGN KEY (pid) REFERENCES Product(pid)
);
```

Here are the examples of the tables that are instances of the above relations.

Product

pid	name	category
1	Laptop	Electronics
2	Desk Chair	Furniture
3	Coffee Maker	Appliances
4	Notebook	Stationery
5	Headphones	Electronics
6	Desk Lamp	Furniture
7	Water Bottle	Accessories

Customer

cid	name	region
1	Alice Brown	West
2	Bob Smith	East
3	Carla Torres	North
4	Daniel Kim	South
5	Eva Johnson	Central

Sales

sale_id	cid	pid	quantity
1	1	1	2
2	1	4	5
3	2	3	1
4	2	7	3
5	3	2	2
6	3	5	1
7	4	1	1
8	4	6	2
9	5	3	4
10	5	7	2

Queries

Query 1.

```
select P.pid, P.category
from Product P, Sales S
where P.name = 'Notebook'
and    P.pid=S.pid
and    S.quantity>10;
```

Query 2.

```
select C.name, P.name, S.quantity
from Product P, Sales S, Customer C
where C.cid=S.cid
and    P.pid=S.pid
and    C.region='West';
```

Exercises

Exercise 1.

How to vectorize Query 1? Sketch the movement of data and the vectorized operations used for the implementation of Query 1.

Exercise 2.

How to approach the vectorization of Query 2? Give an abstract description of the vectorized implementation of Query 2.

Exercise 3.**Descriptive and explanatory questions**

1. Describe the SIMD model of CPU instructions.
2. Assume you are using the programming language C or C++. How to implement vectorized code?
3. Describe the SIMD operations provided by AVX-512.
4. Present some principles employed in the vectorization of DBMS algorithms.
5. Sketch the main ideas for the vectorization of the selection scans.
6. Describe the vector refill algorithms. Why and how is this technique used in DBMS algorithms?
7. What is the relaxed operator fusion?
8. Describe how to vectorize the probing phase of the hash join.